DATA MINING
LAB EXAM

Complete practical answers using WEKA

Experiments covered	8 practical experiments
Software	WEKA Explorer
Includes	Aim, datasets, step-by-step procedures, formulas, expected outputs, results and viva questions

General steps to start WEKA

1. Open WEKA.
2. Select **Explorer**.
3. Use the **Preprocess** tab to load the required ARFF dataset using **Open file**.
4. Choose the appropriate tab (Preprocess, Select attributes, Classify or Cluster) for the experiment.

1. Data Cleaning and Handling Missing Values

Aim: To identify and handle missing values in a dataset using WEKA by applying replace, remove and imputation operations.

Dataset: weather.nominal.arff

Procedure

1. Open WEKA Explorer.
2. Go to the **Preprocess** tab.
3. Click **Open file** and select weather.nominal.arff.
4. Observe the attributes and their values. Missing values are represented by a question mark (?).
5. To remove instances based on missing or selected values, click **Filter** and select **unsupervised** → **instance** → **RemoveWithValues**. Configure the filter if required, then click **Apply**.
6. To replace missing values, select **unsupervised** → **attribute** → **ReplaceMissingValues** and click **Apply**.
7. WEKA replaces missing numeric values with the mean and nominal values with the mode.
8. Observe the updated dataset and missing-value count.

Expected output

• Missing values are identified.

• Instances can be removed using an instance filter.

• Missing values are replaced using suitable statistical values.

Result: Successfully performed data cleaning and handled missing values using WEKA.

2. Data Normalization and Discretization

Aim: To normalize numerical attributes using Min-Max and Z-score techniques and convert numerical attributes into nominal attributes using discretization.

Dataset: iris.arff

A. Min-Max normalization

1. Open WEKA Explorer and load iris.arff in the Preprocess tab.
2. Click **Filter** and select **unsupervised** → **attribute** → **Normalize**.
3. Click **Apply**. Numeric values are scaled between 0 and 1.

Formula: X' = (X – Xmin) / (Xmax – Xmin)

B. Z-score normalization

1. Click Filter and select **unsupervised** → **attribute** → **Standardize**.
2. Click **Apply**. Values are transformed using the mean and standard deviation.

Formula: $Z = (X - \mu) / \sigma$

• X = original value

• μ = mean

• σ = standard deviation

C. Discretization

1. Reload the original iris.arff dataset.
2. Select **Filter** → **unsupervised** → **attribute** → **Discretize**.

3. Configure the number of bins (for example, 10).
4. Click **Apply**. Numerical attributes are converted into nominal intervals.

Expected output

- Normalize: values between 0 and 1.
- Standardize: values with mean approximately 0 and standard deviation approximately 1.
- Discretize: numeric values converted into categorical intervals.

Result: Successfully performed normalization and discretization using WEKA.

3. Attribute Selection and Dimensionality Reduction

Aim: To select important attributes using InfoGain, GainRatio and Ranker evaluators, and reduce dimensionality using Principal Component Analysis (PCA).

Dataset: iris.arff

A. Attribute selection using InfoGain

1. Open WEKA Explorer and load iris.arff.
2. Select the **Select attributes** tab.
3. Under Attribute Evaluator, choose **InfoGainAttributeEval**.
4. Under Search Method, select **Ranker**.
5. Click **Start** and observe the information gain values and attribute ranking.

B. Attribute selection using GainRatio

1. Select **GainRatioAttributeEval** under Attribute Evaluator.
2. Keep Search Method as **Ranker**.
3. Click **Start** and observe the ranking of attributes.

C. Dimensionality reduction using PCA

1. Go to the **Preprocess** tab and load iris.arff.
2. Click **Filter** → **unsupervised** → **attribute** → **PrincipalComponents**.
3. Configure the number of components if required.
4. Click **Apply** and observe the reduced number of attributes.

Expected output

- InfoGain and GainRatio rank attributes according to their importance.
- Ranker displays attributes in descending order.
- PCA transforms the original attributes into fewer principal components while retaining as much variance as possible.

Result: Successfully performed attribute selection and dimensionality reduction using WEKA.

4. Classification Using ID3 Algorithm (J48 in WEKA)

Aim: To perform classification using the J48 decision tree algorithm and evaluate its accuracy using 10-fold cross-validation.

Dataset: iris.arff

Procedure

1. Open WEKA Explorer and load iris.arff in the Preprocess tab.

2. Ensure the class attribute is selected as **class** (species).
3. Go to the **Classify** tab and click **Choose**.
4. Select **trees** → **J48**.
5. Under Test Options, select **Cross-validation** and set the number of folds to **10**.
6. Click **Start** and observe the classification results.

To view the decision tree

1. Right-click the completed result in the Result list.
2. Select **Visualize tree** and observe the decision tree structure.

Accuracy formula: Accuracy = (Correctly classified / Total instances) × 100

Confusion matrix: A confusion matrix compares actual classes with predicted classes.

	Predicted positive	Predicted negative
Actual positive	TP	FN
Actual negative	FP	TN

Expected output

• Correctly Classified Instances and Incorrectly Classified Instances.

• Accuracy percentage, confusion matrix and detailed accuracy by class.

• Decision tree structure.

Result: Successfully classified the iris dataset using the J48 decision tree algorithm and evaluated its performance using 10-fold cross-validation.

5. Classification Using Naive Bayes

Aim: To classify datasets using the Naive Bayes algorithm and compare its performance with the decision tree algorithm.

Dataset: weather.nominal.arff or diabetes.arff

A. Classification using Naive Bayes

1. Open WEKA Explorer and load weather.nominal.arff.
2. Select the appropriate class attribute.
3. Go to the **Classify** tab and click **Choose**.
4. Select **bayes** → **NaiveBayes**.
5. Under Test Options, select **Cross-validation** and set folds to **10**.
6. Click **Start** and observe accuracy and the confusion matrix.

B. Classification using decision tree

1. Keep the same dataset.
2. Select **trees** → **J48**.
3. Select 10-fold cross-validation and click **Start**.
4. Observe the accuracy and confusion matrix.

C. Compare the results

Evaluation	Naive Bayes	J48
Correctly classified	Record output	Record output
Incorrectly classified	Record output	Record output
Accuracy	Record output	Record output
Confusion matrix	Record output	Record output

Naive Bayes formula: $P(C|X) = [P(X|C) \times P(C)] / P(X)$

- $P(C|X)$ = probability of class given the data.
- $P(X|C)$ = probability of data given the class.
- $P(C)$ = prior probability of class.
- $P(X)$ = probability of data.

Result: Successfully performed classification using Naive Bayes and compared its performance with the J48 decision tree algorithm.

6. Classification Using k-Nearest Neighbors (IBk)

Aim: To perform classification using the k-Nearest Neighbors algorithm and analyze the effect of different k values on accuracy.

Dataset: iris.arff

Procedure

1. Open WEKA Explorer and load iris.arff.
2. Go to the **Classify** tab and click **Choose**.
3. Select **lazy** → **IBk**.
4. Click the IBk algorithm name to configure its properties.
5. Set the KNN value to **1**.
6. Under Test Options, select **Cross-validation** and set folds to **10**.
7. Click **Start** and record the accuracy.
8. Repeat by changing K values to **3, 5 and 7**.
9. Compare the accuracy for different K values.

Observation table

K value	Accuracy
1	Record output
3	Record output
5	Record output
7	Record output

Working principle

1. Select a value of K.
2. Calculate the distance between the test instance and training instances.
3. Identify the K nearest neighbors.

4. Count the class labels of the neighbors.
5. Assign the majority class to the test instance.

Euclidean distance formula: $d(x,y) = \sqrt{[\Sigma (xi - yi)^2]}$

Expected output

- Classification accuracy for different K values.
- Confusion matrix.
- Correctly and incorrectly classified instances.

Result: Successfully performed classification using the IBk algorithm and analyzed the effect of different K values on classification accuracy.

7. Classification Using Support Vector Machine (SVM)

Aim: To perform classification using the Support Vector Machine algorithm with SMO and evaluate its performance using accuracy and ROC curves.

Dataset: diabetes.arff

A. Classification using SMO

1. Open WEKA Explorer and load diabetes.arff.
2. Select the appropriate class attribute.
3. Go to the **Classify** tab and click **Choose**.
4. Select **functions** → **SMO**.
5. Under Test Options, select **Cross-validation** and set folds to **10**.
6. Click **Start** and observe the classification results and accuracy.

B. Applying Sequential Minimal Optimization

1. Click the SMO algorithm name.
2. Observe its configuration properties.
3. Keep the default settings for the basic experiment.
4. SMO trains the Support Vector Machine by optimizing the model using small subsets of training examples.

C. Generating ROC curve

1. After classification, right-click the result in the Result list.
2. Select **Visualize threshold curve**.
3. Choose the positive class and observe the ROC curve.

ROC: Receiver Operating Characteristic.

- X-axis: False Positive Rate (FPR).
- Y-axis: True Positive Rate (TPR).

TPR formula: $TPR = TP / (TP + FN)$

FPR formula: $FPR = FP / (FP + TN)$

Expected output

- Classification accuracy and confusion matrix.
- Correctly classified instances.
- ROC curve and ROC Area (AUC).

Result: Successfully performed classification using the SVM algorithm with SMO and evaluated its performance using accuracy and ROC curves.

8. Clustering Using K-Means Algorithm

Aim: To perform clustering using the K-Means algorithm and visualize the clusters by comparing them with the actual class labels.

Dataset: iris.arff or bank-data.arff

A. Applying K-Means clustering

1. Open WEKA Explorer and load iris.arff.
2. Go to the **Cluster** tab and click **Choose**.
3. Select **clusterers** → **SimpleKMeans**.
4. Click the algorithm name to configure its properties.
5. Set the number of clusters (NumClusters) to **3**.
6. Under Cluster Mode, select **Classes to clusters evaluation**.
7. Select the actual class attribute (class) for comparison.
8. Click **Start**.

B. Visualizing clusters

1. After clustering, right-click the result in the Result list.
2. Select **Visualize cluster assignments**.
3. Choose the attributes for the X-axis and Y-axis.
4. Observe the different clusters using different colors.
5. Compare the generated clusters with the actual class labels.

Working principle of K-Means

1. Select K initial cluster centers.
2. Calculate the distance between each data point and cluster centers.
3. Assign each data point to its nearest cluster.
4. Recalculate the cluster centers.
5. Repeat until cluster assignments stop changing or convergence is reached.

K-Means objective function: $J = \Sigma$ (over clusters j) Σ (for xi in Cj) $||xi - \mu j||^2$

- K = number of clusters.
- Cj = cluster.
- μj = cluster centroid.
- xi = data point.

Expected output

- Number of clusters and cluster centroids.
- Number of instances in each cluster.
- Within-cluster sum of squared errors (SSE).
- Cluster visualization and comparison with actual class labels.

Result: Successfully performed K-Means clustering and visualized the clusters by comparing them with actual class labels.

Important WEKA Practical Exam Notes

- **Experiment 4:** J48 is WEKA's implementation of the C4.5 decision tree algorithm, not the original ID3 algorithm. If the question explicitly says J48, use J48.
- **Experiment 1:** The standard weather.nominal.arff dataset may not contain missing values. If so, use a dataset containing ? values or introduce missing values before applying the filters.
- **Experiment 5:** Use the same dataset and 10-fold cross-validation settings for both Naive Bayes and J48 to make a fair comparison.
- **Experiment 6:** Change only K when comparing IBk results. Keep other settings identical.
- **Experiment 8:** K-Means does not use actual class labels to create clusters. Class labels are used afterward to evaluate the clusters.

Quick Revision: WEKA Algorithms and Filters

Algorithm / Technique	Purpose
ReplaceMissingValues	Replaces missing values with mean or mode
RemoveWithValues	Removes instances based on selected attribute values
Normalize	Scales values between 0 and 1
Standardize	Converts values into Z-scores
Discretize	Converts numeric values into nominal intervals
InfoGain	Measures information gain of attributes
GainRatio	Measures gain relative to split information
PCA	Reduces dimensionality
J48	Decision tree classification
NaiveBayes	Probabilistic classification
IBk	K-Nearest Neighbors classification
SMO	Support Vector Machine training algorithm
SimpleKMeans	Groups data into K clusters

Important Viva Questions and Answers

1. What is WEKA?

WEKA is an open-source data mining and machine learning software used for data preprocessing, classification, clustering and visualization.

2. What is an ARFF file?

ARFF stands for Attribute-Relation File Format. It stores datasets with attribute names, data types and data values.

3. What is missing data?

Missing data refers to values that are not available in a dataset. It is represented by ? in ARFF files.

4. What is normalization?

Normalization is the process of scaling numerical values into a specific range, usually 0 to 1.

5. What is PCA?

PCA stands for Principal Component Analysis. It reduces the number of dimensions while preserving as much information (variance) as possible.

6. What is classification?

Classification is a supervised learning technique that assigns data to predefined categories or classes.

7. What is clustering?

Clustering is an unsupervised learning technique that groups similar data points together.

8. What is a confusion matrix?

A confusion matrix is a table used to compare actual and predicted classifications.

9. What is cross-validation?

Cross-validation is a technique used to evaluate model performance by dividing data into multiple training and testing sets.

10. What is 10-fold cross-validation?

The dataset is divided into 10 parts. Nine parts are used for training and one part for testing, repeating the process 10 times.

11. What is K in KNN?

K represents the number of nearest neighbors considered when classifying a data point.

12. What is SVM?

Support Vector Machine is a supervised learning algorithm that finds an optimal boundary (hyperplane) to separate different classes.

13. What is ROC?

ROC stands for Receiver Operating Characteristic. It is a graph showing the relationship between the True Positive Rate and False Positive Rate.

14. What is AUC?

AUC stands for Area Under the Curve. A higher AUC generally indicates better classification performance.

15. What is K-Means?

K-Means is an unsupervised clustering algorithm that divides data into K groups based on similarity.